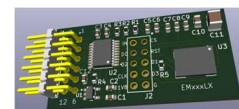


PMOD Background



Digilent designs and supports a broad portfolio of data acquisition, data logging, and development tools for FPGA- and SoC-based systems. To simplify peripheral expansion, Digilent created the PMOD (Peripheral Module) standard—a low-cost, plug-in board ecosystem that adds common functions through a consistent connector and pinout. The Digilent PMOD website provides product details as well as the official PMOD specification.

[Reference: Digilent PMOD documentation](#)

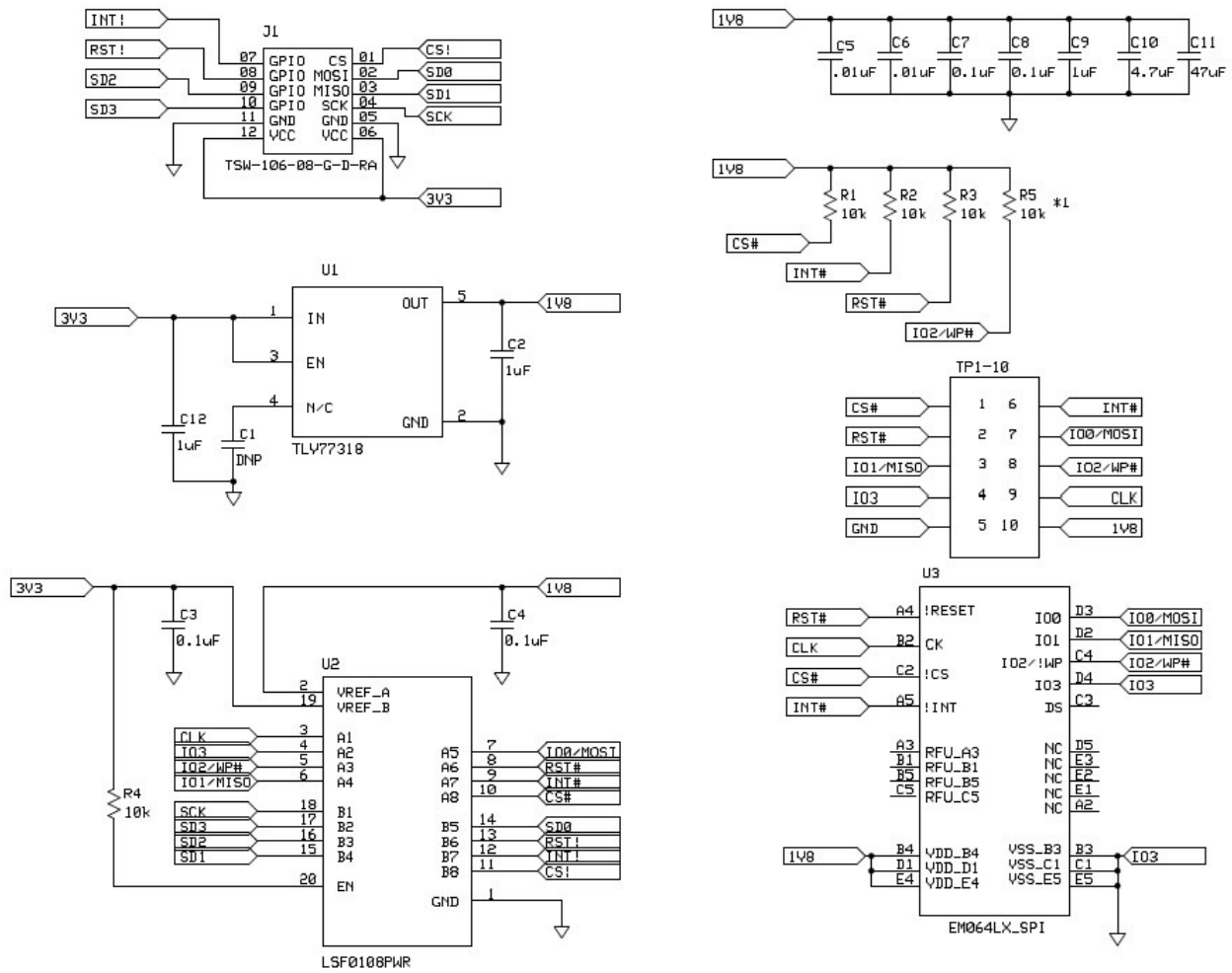
EVERSPIN EM064LX PMOD

Everspin has created a 12-pin PMOD compatible memory module based on its EM064LX MRAM IC. This PMOD is similar to other flash storage solutions. The EM064LX PMOD can operate in single, dual, or quad SPI modes and supports the features of xSPI. The PMOD is supplied in single SPI mode but can be configured as needed by the user. The PMOD specification supports a 3.3V power supply rail. Since the EM064LX IC is based on a 1.8V VDD power rail, this PMOD design includes an LDO and appropriate auto-sensing level shifting for data and control lines. The EM064LX IC itself has a density of 64 Mb and is based on Everspins MRAM technology. You can find more information about the EM064LX on Everspins website and GitHub repository.

Figure 1 below shows the schematic for the EM064LX PMOD. J1 is the PMOD connector and it follows the Digilent Type 2A pinout (see section 5.2 “SPI” in the Digilent specification).

Pin Number	Function	Pin Number	Function
1	CS!	7	INT!
2	SD0	8	RST!
3	SD1	9	SD2
4	SCK	10	SD3
5	GND	11	GND
6	3V3	12	3V3

Table 1: Pins and Functions for Everspin EM064LX PMOD



Notes: #1 For single SPI R5 = 10 kohm, for dual or quad SPI R5 = DNP.

#2 RST#, CLK, CS#, INT#, IO0/MOST, IO1/MISO, IO2/WP#, IO3 are all at 1.8V LVCMOS levels.

Figure 1: Schematic for Everspin EM064LX PMOD

PMOD USAGE

The EM064LX PMOD plugs into any compatible host platform, including SoC- and FPGA-based development environments, providing a simple path to add nonvolatile MRAM to a design. Integrating this module allows designers to leverage MRAM technology to meet system storage requirements while maintaining a standard PMOD interface.

CONCLUSION

Everspin is the leading supplier of persistent MRAM solutions, delivering nonvolatile memory that combines SRAM-like speed with flash-like data retention—without the erase cycles, wear leveling, or write-latency uncertainty of traditional flash devices. Built on Everspin's STT-MRAM technology and JEDEC xSPI interfaces, the EM064LX family enables fast, byte-level writes with deterministic performance and effectively unlimited endurance, making it an ideal fit for data logging, configuration, and mission-critical embedded storage. The EM064LX PMOD makes evaluation simple: plug it into a standard PMOD header on an FPGA or SoC platform, bring up the interface in single/dual/quad SPI as needed, and start validating MRAM benefits in your system within minutes.

References

- "Digilent PMOD Interface Specification" (Last update: October 28, 2020).
[https://digilent.com/reference/ media/reference/pmod/pmod-interface-specification-1 3 1.pdf](https://digilent.com/reference/media/reference/pmod/pmod-interface-specification-131.pdf)
- Everspin Website, EM064LX Product page.
<https://www.everspin.com/family/em064lxo?npath=3843>
- Everspin GitHub: <https://github.com/EVERSPINTECHNOLOGIES>

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